Angelo **Saporito**

MSc Data Science, LSE · Applied Economics & Statistics · US Citizen

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Summary

Applied economist and quantitative researcher completing an MSc in Data Science at the LSE, built on undergraduate degrees in economics and mathematics. Nineteen months as an economics research assistant producing causal policy estimates on federal microdata, and a working paper on monetary policy rules under state uncertainty. My work returns to one question across projects: what a measurement or estimation pipeline discards, and how that shapes the conclusions drawn from it. Seeking a two-year research assistantship as preparation for a PhD in economics. US citizen; available from October 2026.

Research Experience

Economic Policy Analyses of Pennsylvania's Act 39 and Clean Slate Legislation

Dept. of Economics, WCU

Research Assistant to Simon Condliffe, Ph.D. and Matt Saboe, Ph.D. *Dec 2022 – Jun 2024*

- Estimated the employment and output effects of Act 39 on Pennsylvania's beer and wine industry using a regression-discontinuity design with IMPLAN input–output modelling, on Bureau of Labor Statistics and Department of Labor microdata.
- Analysed the Clean Slate Act, which automates the sealing of criminal records, and its relationship to employment, housing, and education access using ordinal logistic and categorical-data methods. Associational rather than causal, and reported as such.
- Built and maintained the data pipeline in SQL and SAS, co-authored both reports, and presented findings at two conferences, one national and one regional.

Belief-State Structure in Reinforcement Learning for Monetary Policy

LSE

Working Paper *2026*

- Asked what happens to a learned policy rule when the policymaker observes a forecast rather than the state. Central-bank agents are trained in a three-equation New Keynesian economy with an analytical Taylor-rule benchmark, observing a belief supplied by a forecaster-informed noise model. **Code:** github.com/angelo-saporito24/Macro-forecasting
- **Finding:** interposing a learned belief between the state and the policy destroys the determinacy property of the rule. The long-run inflation response falls from a benchmark of 1.66 to between 0.011 and 0.025 across all twelve partially observed runs, and is statistically indistinguishable from zero ($p = 0.22$–0.52), while all twenty-four runs in which a response is present are significant at $p < 10^{-8}$. Consistent with attenuation under signal noise roughly five times the simulated dispersion of inflation.
- The calibration contrast the design set out to measure is reported as not identified, since both conditions suffer identical first-moment corruption. Reported as a negative result with limitations and an unstable control disclosed.

Partially Identified: Privacy Suppression and the Automation–Augmentation Measure in the Anthropic Economic Index

Working Paper *2026*

- Shows that a widely adopted observed-usage measure of AI exposure is partially identified; the index itself is cited in Federal Reserve Board staff research. The Anthropic Economic Index suppresses any cell holding fewer than fifteen conversations from fewer than five accounts, which censors the data on two margins: modes within observed tasks, and whole tasks within occupations. Bounding both, 42 of 775 six-digit SOC occupations, or 5.4%, have an identified set narrower than 0.10 on at least five observed tasks, and sixteen of 22 major groups contain none.
- Suppression removes only 6.38% of classified conversation volume, yet the loss of identification is far larger, because suppression concentrates in thin cells and most tasks are thin. Identification proves to be a property of data density rather than of the work: log classified volume alone explains 76% of the variation in joint bound width, and 21 occupational dummies add 1.2 percentage points on top of volume and coverage.
- Documents a near-cancellation in which suppression biases a reconstruction of the published global automation share upward 0.55pp while a commonly discarded pseudo-task biases it downward 0.47pp, so a pipeline that has failed twice appears to validate. Assesses the consequences for occupation-level wage regressions, published regional exposure indices, and the practice of recoding fully suppressed tasks to manual. Reproducible R pipeline with assertion checks and a censoring sensitivity sweep. **Code:** github.com/angelo-saporito24/Partially-Identified

Modelling Tail Risk and Asymmetry in Energy Spread Trading

Dept. of Statistics, LSE

MSc Capstone, industry project *Jan – Sep 2026*

- With a European energy-trading firm: modelled tail risk in the ERCOT electricity market using conditional-autoregressive quantile (CAViaR) models of the day-ahead against real-time spread across four hubs, with 13,872 monthly model fits over seventeen delivery months.
- **Finding:** statistical accuracy and economic value can come apart: the most accurate specification lost its calibration in the 2026 holdout and was not an economic winner. Out-of-sample discipline imposed with Diebold–Mariano testing and no-lookahead lags; the two surviving drivers matched grid-economics priors and were used as a validity check.

Ambiguity as Evidence: Uncertainty Propagation in Computational Text Analysis *Dept. of Statistics, LSE*

MSc Project, Bayesian Machine Learning *2026*

- Propagated preprocessing uncertainty through topic models of twelve works of Plato and Aristotle, and decomposed the resulting variance to separate genuine measurement uncertainty from the estimator's own initialisation noise. Rebutted a published claim by holding corpus size fixed by construction. **Code:** github.com/angelo-saporito24/ST451_project

Education

The London School of Economics and Political Science *London, UK*

MSc Data Science *Sep 2025 – Sep 2026*

- **Distinction expected.** Coursework: Machine Learning & Data Mining, Bayesian Machine Learning, Reinforcement Learning, Deep Learning, Data Analysis & Statistical Methods, Managing & Visualising Data.
- Audited graduate courses, Dept. of Philosophy, Logic and Scientific Method: PH434 Philosophy of Economics; PH456 Rationality & Choice; PH400 Philosophy of Science; PH437 Advanced Logic; PH440 Ethics of Data and AI.

West Chester University of Pennsylvania *West Chester, PA*

B.S. Economics; B.S. Mathematics (concentrations in Statistics and Mathematical Finance) *Aug 2019 – May 2024*

- **GPA 3.78/4.00** across 204 credit hours. *Summa Cum Laude*; Dean's List in eight semesters; Omicron Delta Epsilon. Minor in Political Science.
- **Economics:** Intermediate Macroeconomics; Intermediate Microeconomics; International Economics; Research Methods in Business and Economics; Business Analytics I and II; Business Analytics Using R; Principles of Macroeconomics and Microeconomics. Related: International Political Economy; Independent Studies in Political Science.
- **Statistics and quantitative methods:** Mathematical Statistics I and II; Applied Probability; Applied Statistics; Experimental Design; Basics of Statistical Learning; Topics in Advanced Statistics; Statistical Computing and Data Management; Capstone Course in Statistics.
- **Mathematics:** Calculus I–III; Linear Algebra; Differential Equations; Real Analysis I; Applied Analysis I; Topology; Numerical Analysis I; Optimization Techniques; Financial Calculus; Mathematics of Finance; Mathematical Physics.

Methods and Software

- **Econometrics**: causal inference and regression discontinuity; input–output (IMPLAN) modelling; time-series and panel data; quantile regression (CAViaR, VaR); ordinal logistic and categorical data; maximum likelihood; bootstrap and Monte Carlo; heteroskedasticity-robust inference.
- **Statistical software**: Python (advanced) — pandas, NumPy, SciPy, statsmodels, scikit-learn, PyTorch, numba; R (tidyverse, caret); SAS (Macro, SQL); SQL; MATLAB; LaTeX; Git.
- **Machine learning**: reinforcement learning (PPO, SAC, DQN, offline RL); Bayesian inference (variational Bayes, MCMC, Gaussian processes); deep learning; probabilistic forecasting and calibration (CRPS, proper scoring rules).
- **Data engineering**: REST APIs and web scraping; structured extraction from JSON and PDF; large-scale data pipelines; reproducible research repositories with automated testing.

Conference Presentations

- Murphy, J., **Saporito, A.**, Condliffe, S., Saboe, M., & Ocean, M. (2023). *Living with a Criminal Record in Pennsylvania and the Impact of Clean Slate.* 78th Annual Meeting of the American Society of Criminology, Philadelphia, PA.
- Murphy, J., **Saporito, A.**, Condliffe, S., & Saboe, M. (2023). *Living with a Criminal Record in Pennsylvania.* 46th Annual Conference of the Northeastern Association of Criminal Justice Sciences, Williamsport, PA.

Awards

- **Mathematics Endowment Award**, West Chester University (2023), for excellence in the mathematics curriculum and for tutoring undergraduate peers.
- **Graduate Datathon, 2nd Place**, WCU–Fulton Bank (2023), sole undergraduate among graduate teams; modelled 5M+ transaction records and presented recommendations to bank executives.